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Title & Objective

Research Question:

How do income, education, race/ethnicity, and rurality relate to heart disease mortality across Texas counties (2022)?

Audience:

Public health stakeholders



Why This Matters



Leading Cause

Heart disease = leading cause of death in Texas



Disparities

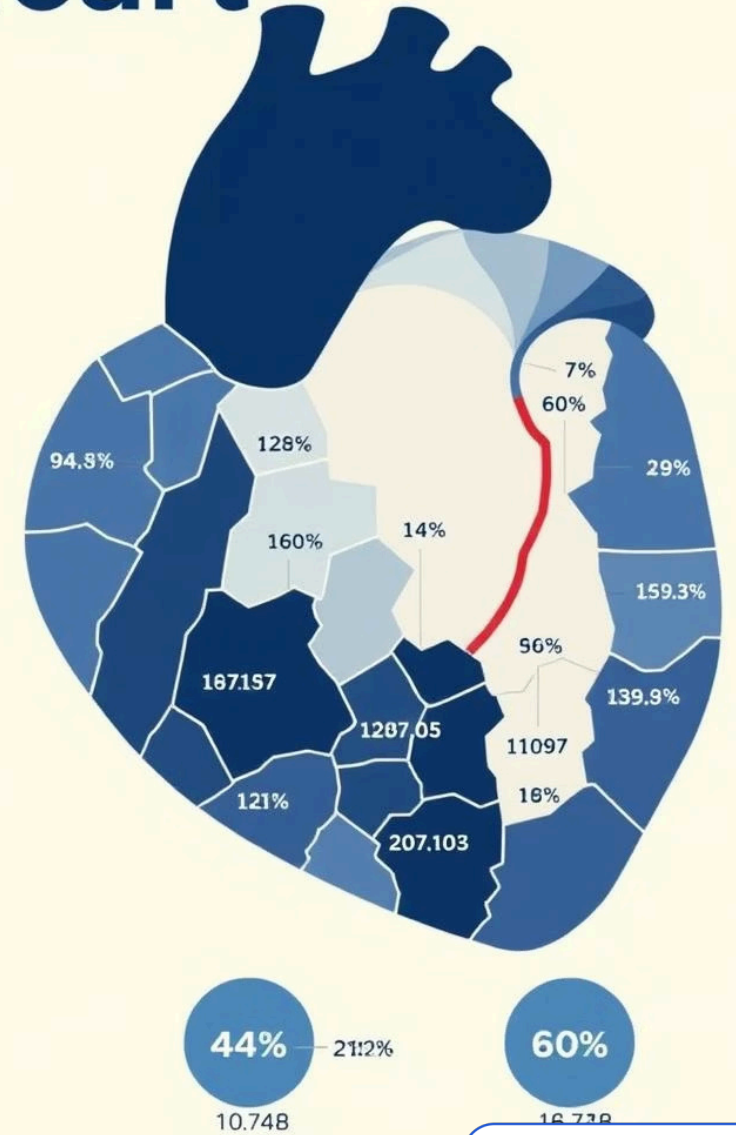
Rural + low-income areas disproportionately impacted



Social Factors

Social determinants are key to prevention & policy

Heart



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Data & Methods

Dataset

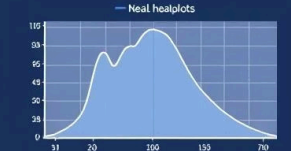
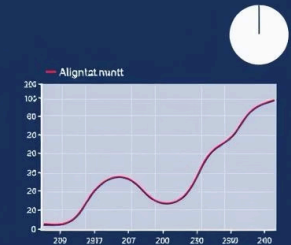
County-level mortality +
socioeconomic indicators
(2022)

Methods

Data cleaning, EDA,
correlations, OLS regression

Tools

Boxplots, heatmaps,
regression outputs



Key Finding 1: Income & Education

- Higher income → lower mortality
- Education improves outcomes, especially in urban areas
- Evidence: Boxplot of mortality by income quartile



Income

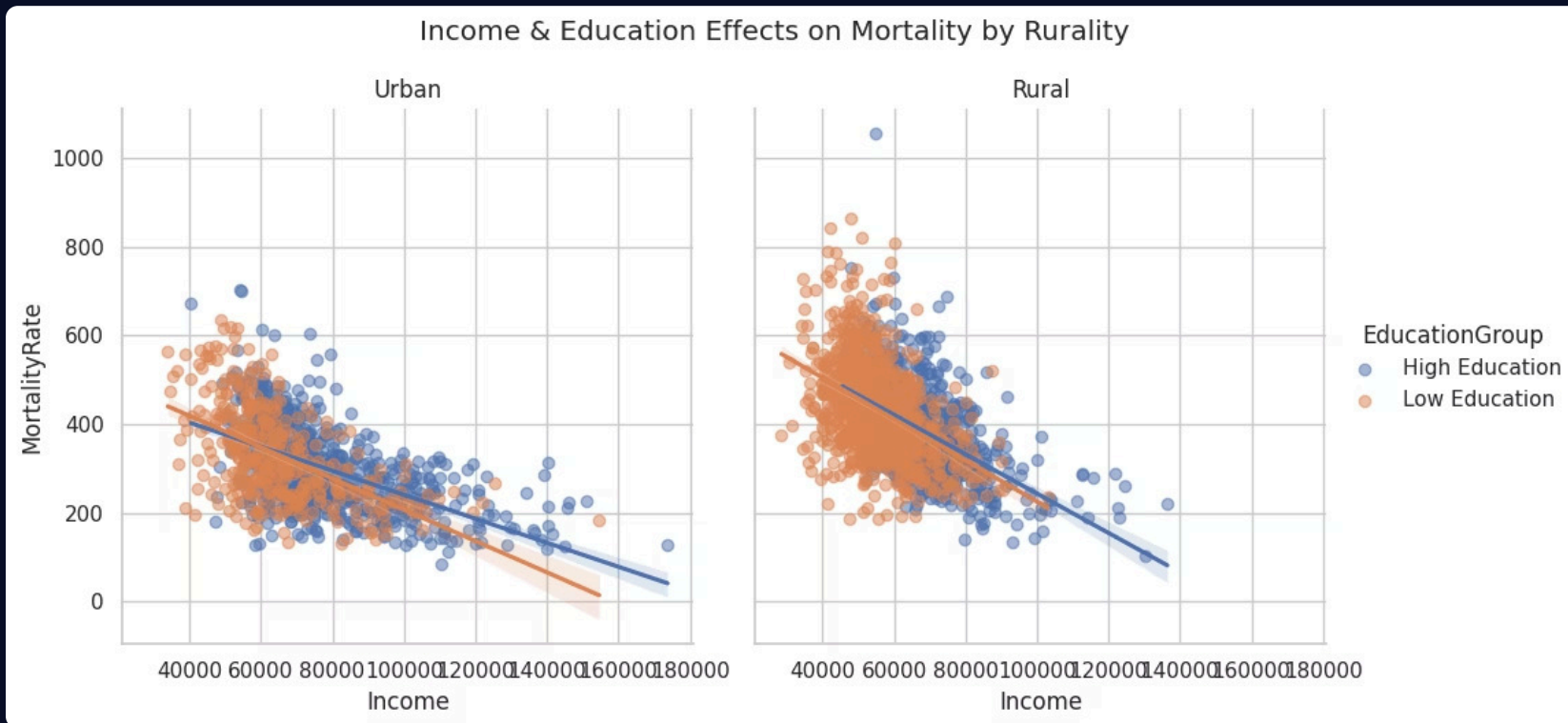
Higher income counties show significantly lower heart disease mortality rates



Education

Educational attainment correlates with better health outcomes

Income & Education effects by rurality



Key Finding 2: Race & Ethnicity

Hispanic Population

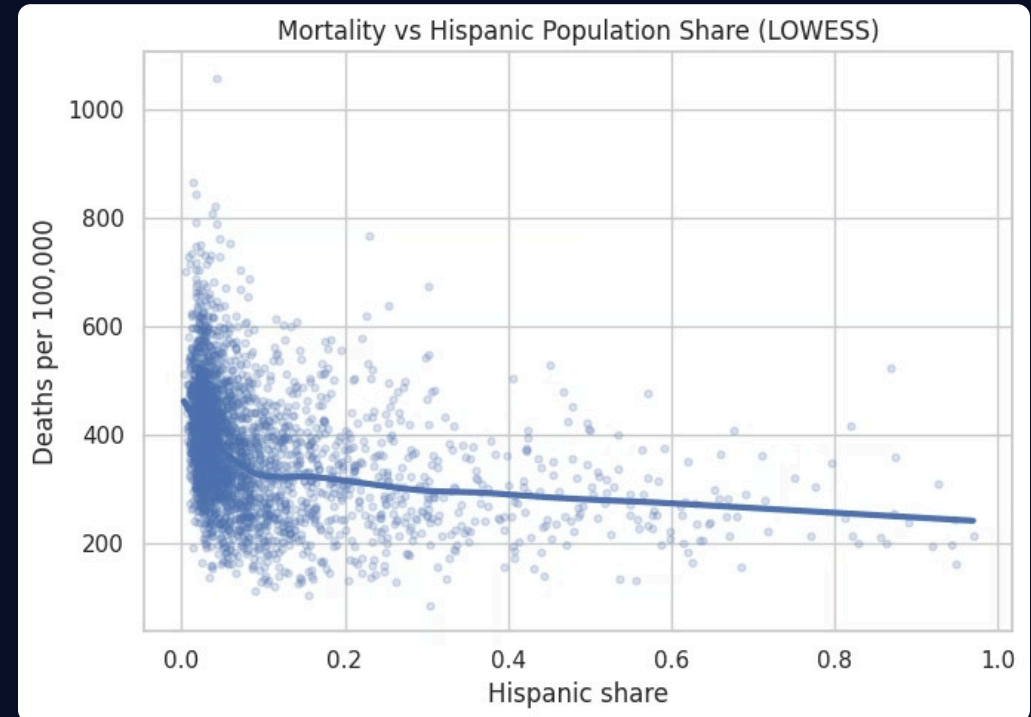
Hispanic population share linked to lower mortality

Black Population

Black population shows weaker statistical link in model

Evidence

LOWESS plot of mortality vs Hispanic share

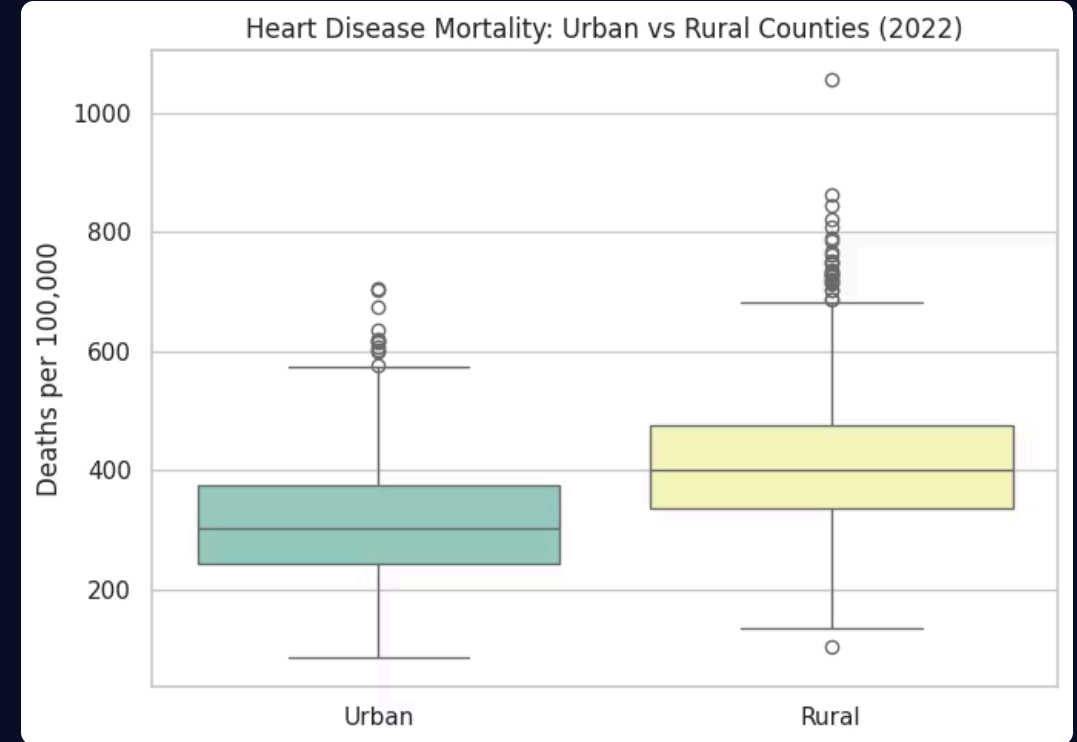


Key Finding 3: Rurality

Rural counties have significantly higher mortality

Regression confirms rurality positively predicts mortality

Evidence: Boxplot of mortality by rural vs urban



📌 Rural Health Disparities

The data clearly shows that living in a rural county is a significant risk factor for heart disease mortality, even when controlling for other socioeconomic factors.



Limitations & Next Steps

Data Limitations

Only 2022 snapshot (no time trend)

Limited variables: No behavioral/clinical risk factors

1

2

Next Steps

Add longitudinal data + healthcare access measures

Conclusion & Policy Implications

Strongest Predictors

Income, education, and rurality

Policy Recommendations

Improve rural health access +
reduce income/education
disparities

Next Steps

Tailored interventions, community programs

